

AirForce Management System

SQL QUERIES and Screenshots:

Step 1: Create Database

```
CREATE DATABASE nexusair_crisishub;
```

Then connect to it:

```
\c nexusair_crisishub
```

If you're using pgAdmin, simply create the database and open the Query Tool for that database.

Step 2: Create Tables

Aircraft

```
CREATE TABLE Aircraft (  
    AircraftID SERIAL PRIMARY KEY,  
    AircraftType VARCHAR(100),  
    Capacity INT,  
    Status VARCHAR(50)  
);
```

Pilot

```
CREATE TABLE Pilot (  
    PilotID SERIAL PRIMARY KEY,  
    PilotName VARCHAR(100),  
    LicenseNo VARCHAR(50) UNIQUE,  
    Phone VARCHAR(20),  
    Status VARCHAR(50)  
);
```

Passenger

```
CREATE TABLE Passenger (  
    PassengerID SERIAL PRIMARY KEY,  
    FullName VARCHAR(100),  
    Phone VARCHAR(20),  
    Email VARCHAR(100)  
);
```

Staff

```
CREATE TABLE Staff (  
    StaffID SERIAL PRIMARY KEY,  
    StaffName VARCHAR(100),  
    Role VARCHAR(50),  
    Phone VARCHAR(20)  
);
```

Flight

```
CREATE TABLE Flight (  
    FlightID SERIAL PRIMARY KEY,  
    FlightNo VARCHAR(20),  
    Origin VARCHAR(100),  
    Destination VARCHAR(100),  
    DepartureTime TIMESTAMP,  
    ArrivalTime TIMESTAMP,  
    Status VARCHAR(50),  
    AircraftID INT REFERENCES Aircraft(AircraftID)  
);
```

Flight_Assignment

```
CREATE TABLE Flight_Assignment (  
    AssignmentID SERIAL PRIMARY KEY,  
    FlightID INT REFERENCES Flight(FlightID),  
    PilotID INT REFERENCES Pilot(PilotID),  
    Role VARCHAR(50)  
);
```

Booking

```
CREATE TABLE Booking (  
    BookingID SERIAL PRIMARY KEY,  
    PassengerID INT REFERENCES Passenger(PassengerID),  
    FlightID INT REFERENCES Flight(FlightID),  
    BookingDate DATE,  
    Status VARCHAR(50)  
);
```

Rebooking

```
CREATE TABLE Rebooking (  
    RebookingID SERIAL PRIMARY KEY,  
    BookingID INT REFERENCES Booking(BookingID),  
    NewFlightID INT REFERENCES Flight(FlightID),  
    Reason VARCHAR(255),  
    RebookedAt TIMESTAMP  
);
```

Disruption

```
CREATE TABLE Disruption (  
    DisruptionID SERIAL PRIMARY KEY,  
    FlightID INT REFERENCES Flight(FlightID),  
    Type VARCHAR(100),  
    Description TEXT,  
    ReportedAt TIMESTAMP,  
    Status VARCHAR(50)  
);
```

Disruption_Resolution

```
CREATE TABLE Disruption_Resolution (  
    ResolutionID SERIAL PRIMARY KEY,  
    DisruptionID INT REFERENCES Disruption(DisruptionID),  
    ActionTaken TEXT,  
    ResolvedAt TIMESTAMP,
```

```
ResolvedBy INT REFERENCES Staff(StaffID)
);
```

Baggage

```
CREATE TABLE Baggage (
    BaggageID SERIAL PRIMARY KEY,
    PassengerID INT REFERENCES Passenger(PassengerID),
    TagNo VARCHAR(50),
    CheckInFlightID INT REFERENCES Flight(FlightID),
    Status VARCHAR(50)
);
```

Baggage_Misdirect

```
CREATE TABLE Baggage_Misdirect (
    MisdirectID SERIAL PRIMARY KEY,
    BaggageID INT REFERENCES Baggage(BaggageID),
    WrongDestination VARCHAR(100),
    FoundAt VARCHAR(100),
    Status VARCHAR(50)
);
```

Baggage_Resolution

```
CREATE TABLE Baggage_Resolution (
    ResolutionID SERIAL PRIMARY KEY,
    MisdirectID INT REFERENCES Baggage_Misdirect(MisdirectID),
    ActionTaken TEXT,
    DeliveredAt TIMESTAMP,
    DeliveredBy INT REFERENCES Staff(StaffID)
);
```

Alert

```
CREATE TABLE Alert (
    AlertID SERIAL PRIMARY KEY,
    StaffID INT REFERENCES Staff(StaffID),
```

```
Message TEXT,  
CreatedAt TIMESTAMP,  
IsRead BOOLEAN  
);
```

Step 3: Insert Sample Data

Run these before running feature queries.

```
INSERT INTO Aircraft (AircraftType, Capacity, Status)  
VALUES ('Airbus A320',180,'Active');
```

```
INSERT INTO Pilot (PilotName, LicenseNo, Phone, Status)  
VALUES ('Ahmed Ali','LIC001','03001234567','Available');
```

```
INSERT INTO Passenger (FullName, Phone, Email)  
VALUES ('Ali Khan','03111234567','ali@gmail.com');
```

```
INSERT INTO Staff (StaffName, Role, Phone)  
VALUES ('Sara Ahmed','Operations Manager','03211234567');
```

```
INSERT INTO Flight  
(FlightNo, Origin, Destination, DepartureTime, ArrivalTime, Status, AircraftID)  
VALUES  
(  
'PK301',  
'Lahore',  
'Karachi',  
'2026-06-10 08:00:00',  
'2026-06-10 10:00:00',  
'Scheduled',  
1);
```

Step 4: Feature Queries and Screenshots

Feature 1 – Assign Pilot

```
INSERT INTO Flight_Assignment  
(FlightID, PilotID, Role)  
VALUES  
(1,1,'Captain');
```

```
SELECT* FROM pilot
```

OneCompiler

Looking for Classic Editor?

AI

MySQL

Run

Save

Login

queries.sql

```
160 VALUES
161 (1,'Dubai','Karachi Airport','Open');
162
163 INSERT INTO Baggage_Resolution
164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)
165 VALUES
166 (1,
167 'Transferred via courier',
168 CURRENT_TIMESTAMP,
169 1);
170 INSERT INTO Alert
171 (StaffID, Message, CreatedAt, IsRead)
172 VALUES
173 (1,
174 'Flight PK301 delayed due to weather',
175 CURRENT_TIMESTAMP,
176 FALSE);
177
178 SELECT * FROM pilot;
```

I/O

AI Agent

373 ms

STDIN

Input for the program (Optional)

Output:

PilotID	PilotName	LicenseNo	Phone	Status
1	Ahmed Ali	LIC001	03001234567	Available

Feature 2 – Create Booking

```
INSERT INTO Booking
(PassengerID, FlightID, BookingDate, Status)
VALUES
(1,1,CURRENT_DATE,'Confirmed');

SELECT* FROM booking
```

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MySQL

Run

Save

Login

queries.sql

```
160 VALUES
161 (1,'Dubai','Karachi Airport','Open');
162
163 INSERT INTO Baggage_Resolution
164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)
165 VALUES
166 (1,
167 'Transferred via courier',
168 CURRENT_TIMESTAMP,
169 1);
170 INSERT INTO Alert
171 (StaffID, Message, CreatedAt, IsRead)
172 VALUES
173 (1,
174 'Flight PK301 delayed due to weather',
175 CURRENT_TIMESTAMP,
176 FALSE);
177
178 SELECT * FROM Booking;
```

I/O

AI Agent

168 ms

STDIN

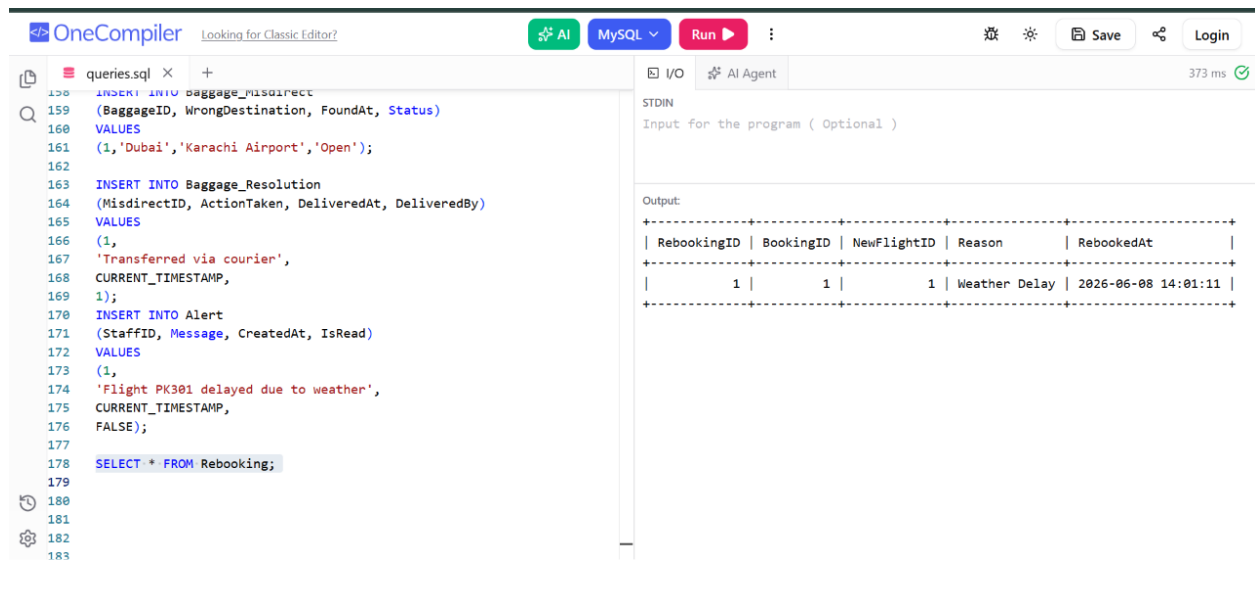
Input for the program (Optional)

Output:

BookingID	PassengerID	FlightID	BookingDate	Status
1	1	1	2026-06-08	Confirmed

Feature 3 – Rebook Passenger

```
INSERT INTO Rebooking  
(BookingID, NewFlightID, Reason, RebookedAt)  
VALUES  
(1,1,'Weather Delay',CURRENT_TIMESTAMP);  
  
SELECT * FROM Rebooking;
```



The screenshot shows the OneCompiler MySQL IDE interface. The left pane displays a SQL query file named 'queries.sql' with the following content:

```
158 INSERT INTO baggage_misdirect  
159 (BaggageID, WrongDestination, FoundAt, Status)  
160 VALUES  
161 (1,'Dubai','Karachi Airport','Open');  
162  
163 INSERT INTO Baggage_Resolution  
164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)  
165 VALUES  
166 (1,  
167 'Transferred via courier',  
168 CURRENT_TIMESTAMP,  
169 1);  
170 INSERT INTO Alert  
171 (StaffID, Message, CreatedAt, IsRead)  
172 VALUES  
173 (1,  
174 'Flight PK301 delayed due to weather',  
175 CURRENT_TIMESTAMP,  
176 FALSE);  
177  
178 SELECT * FROM Rebooking;
```

The right pane shows the output of the query, which is a table with 5 columns: RebookingID, BookingID, NewFlightID, Reason, and RebookedAt. The output is as follows:

RebookingID	BookingID	NewFlightID	Reason	RebookedAt
1	1	1	Weather Delay	2026-06-08 14:01:11

Feature 4 – Report Disruption

```
INSERT INTO Disruption  
(FlightID, Type, Description, ReportedAt, Status)  
VALUES  
(1,  
'Pilot Sick',  
'Pilot reported illness before departure',  
CURRENT_TIMESTAMP,  
'Open');  
  
SELECT * FROM Disruption;
```

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AI

MySQL

Run

🔍

⚙️

💾 Save

🔊

Login

queries.sql

+

159

(BaggageID, WrongDestination, FoundAt, Status)

160

VALUES

161

(1, 'Dubai', 'Karachi Airport', 'Open');

162

163

INSERT INTO Baggage_Resolution

164

(MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)

165

VALUES

166

(1,

167

'Transferred via courier',

168

CURRENT_TIMESTAMP,

169

1);

170

INSERT INTO Alert

171

(StaffID, Message, CreatedAt, IsRead)

172

VALUES

173

(1,

174

'Flight PK301 delayed due to weather',

175

CURRENT_TIMESTAMP,

176

FALSE);

177

178

SELECT * FROM Disruption;

179

180

181

182

183

I/O

AI Agent

401 ms

STDIN

Input for the program (Optional)

Output:

DisruptionID	FlightID	Type	Description
1	1	Pilot Sick	Pilot reported illness before departure

Ready

Light

Wiki

MySQL

Feature 5 – Resolve Disruption

```

INSERT INTO Disruption_Resolution
(DisruptionID, ActionTaken, ResolvedAt, ResolvedBy)
VALUES
(1,
'Replacement pilot assigned',
CURRENT_TIMESTAMP,
1);

```

```

SELECT * FROM Disruption_Resolution;

```

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AI

MySQL

Run

🔍

⚙️

💾 Save

🔊

Login

queries.sql

+

159

(BaggageID, WrongDestination, FoundAt, Status)

160

VALUES

161

(1, 'Dubai', 'Karachi Airport', 'Open');

162

163

INSERT INTO Baggage_Resolution

164

(MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)

165

VALUES

166

(1,

167

'Transferred via courier',

168

CURRENT_TIMESTAMP,

169

1);

170

INSERT INTO Alert

171

(StaffID, Message, CreatedAt, IsRead)

172

VALUES

173

(1,

174

'Flight PK301 delayed due to weather',

175

CURRENT_TIMESTAMP,

176

FALSE);

177

178

SELECT * FROM Disruption_Resolution;

179

180

181

182

183

I/O

AI Agent

382 ms

STDIN

Input for the program (Optional)

Output:

ResolutionID	DisruptionID	ActionTaken	ResolvedAt
1	1	Replacement pilot assigned	2026-06-08 14:02:27

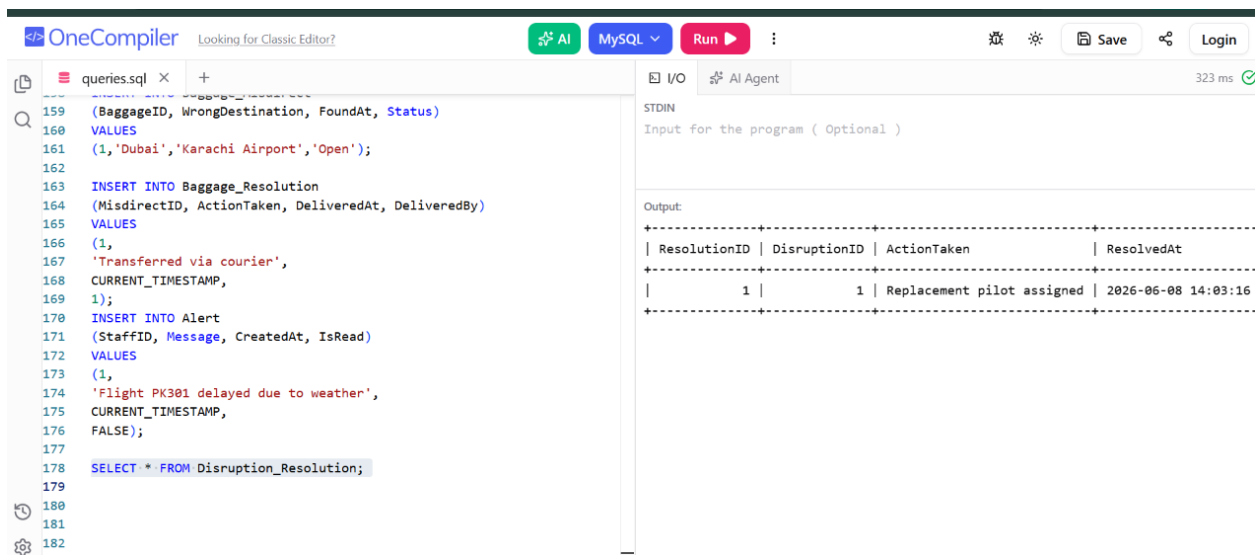
Power & Battery

Energy saver is on

Feature 6 – Add Baggage

```
INSERT INTO Baggage
(PassengerID, TagNo, CheckInFlightID, Status)
VALUES
(1,'BG1001',1,'CheckedIn');

SELECT * FROM Disruption_Resolution;
```



The screenshot shows the OneCompiler MySQL editor interface. The left pane displays a SQL script with line numbers 159 to 182. The script includes three INSERT statements and a SELECT statement. The right pane shows the output of the SELECT statement, which is a table with four columns: ResolutionID, DisruptionID, ActionTaken, and ResolvedAt. The output shows a single row with values 1, 1, 'Replacement pilot assigned', and '2026-06-08 14:03:16'.

```
159 (BaggageID, WrongDestination, FoundAt, Status)
160 VALUES
161 (1,'Dubai','Karachi Airport','Open');
162
163 INSERT INTO Baggage_Resolution
164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)
165 VALUES
166 (1,
167 'Transferred via courier',
168 CURRENT_TIMESTAMP,
169 1);
170 INSERT INTO Alert
171 (StaffID, Message, CreatedAt, IsRead)
172 VALUES
173 (1,
174 'Flight PK301 delayed due to weather',
175 CURRENT_TIMESTAMP,
176 FALSE);
177
178 SELECT * FROM Disruption_Resolution;
179
180
181
182
```

Output:

ResolutionID	DisruptionID	ActionTaken	ResolvedAt
1	1	Replacement pilot assigned	2026-06-08 14:03:16

Feature 7 – Record Misdirected Baggage

```
INSERT INTO Baggage_Misdirect
(BaggageID, WrongDestination, FoundAt, Status)
VALUES
(1,'Dubai','Karachi Airport','Open');

SELECT * FROM Baggage_Misdirect;
```

OneCompiler

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MySQL

Run

Save

Login

queries.sql

+

159 (BaggageID, WrongDestination, FoundAt, Status)
160 VALUES
161 (1, 'Dubai', 'Karachi Airport', 'Open');
162
163 INSERT INTO Baggage_Resolution
164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)
165 VALUES
166 (1,
167 'Transferred via courier',
168 CURRENT_TIMESTAMP,
169 1);
170 INSERT INTO Alert
171 (StaffID, Message, CreatedAt, IsRead)
172 VALUES
173 (1,
174 'Flight PK301 delayed due to weather',
175 CURRENT_TIMESTAMP,
176 FALSE);
177 SELECT * FROM Baggage_Misdirect;
178
179
180
181
182
183

I/O

AI Agent

254 ms

STDIN
Input for the program (Optional)

Output:

MisdirectID	BaggageID	WrongDestination	FoundAt	Status
1	1	Dubai	Karachi Airport	Open

Feature 8 – Resolve Baggage Issue

```
INSERT INTO Baggage_Resolution
(MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)
VALUES
(1,
'Transferred via courier',
CURRENT_TIMESTAMP,
1);

SELECT * FROM Baggage_Resolution;
```

OneCompiler

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AI

MySQL

Run

Save

Login

queries.sql

159 (BaggageID, WrongDestination, FoundAt, Status)

160 VALUES

161 (1, 'Dubai', 'Karachi Airport', 'Open');

162

163 INSERT INTO Baggage_Resolution

164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)

165 VALUES

166 (1,

167 'Transferred via courier',

168 CURRENT_TIMESTAMP,

169 1);

170 INSERT INTO Alert

171 (StaffID, Message, CreatedAt, IsRead)

172 VALUES

173 (1,

174 'Flight PK301 delayed due to weather',

175 CURRENT_TIMESTAMP,

176 FALSE);

177 SELECT * FROM Baggage_Resolution;

178

179

180

181

182

183

I/O

AI Agent

387 ms

STDIN

Input for the program (Optional)

Output:

ResolutionID	MisdirectID	ActionTaken	DeliveredAt	Del
1	1	Transferred via courier	2026-06-08 14:04:24	

Feature 9 – Generate Alert

INSERT INTO Alert
(StaffID, Message, CreatedAt, IsRead)
VALUES
(1,
'Flight PK301 delayed due to weather',
CURRENT_TIMESTAMP,
FALSE);

SELECT * FROM Alert;

OneCompiler

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AI

MySQL

Run

Save

Login

queries.sql

159 (BaggageID, WrongDestination, FoundAt, Status)

160 VALUES

161 (1, 'Dubai', 'Karachi Airport', 'Open');

162

163 INSERT INTO Baggage_Resolution

164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)

165 VALUES

166 (1,

167 'Transferred via courier',

168 CURRENT_TIMESTAMP,

169 1);

170 INSERT INTO Alert

171 (StaffID, Message, CreatedAt, IsRead)

172 VALUES

173 (1,

174 'Flight PK301 delayed due to weather',

175 CURRENT_TIMESTAMP,

176 FALSE);

177 SELECT * FROM Alert;

178

179

180

181

182

I/O

AI Agent

268 ms

STDIN

Input for the program (Optional)

Output:

AlertID	StaffID	Message	CreatedAt
1	1	Flight PK301 delayed due to weather	2026-06-08 14:05:09

Feature 10 – View Open Disruptions

```
SELECT *  
FROM Disruption  
WHERE Status='Open';
```

OneCompiler | Looking for Classic Editor? | AI | MySQL | Run | Save | Login

queries.sql | 344 ms

```
159 (BaggageID, WrongDestination, FoundAt, Status)  
160 VALUES  
161 (1, 'Dubai', 'Karachi Airport', 'Open');  
162  
163 INSERT INTO Baggage_Resolution  
164 (MisdirectID, ActionTaken, DeliveredAt, DeliveredBy)  
165 VALUES  
166 (1,  
167 'Transferred via courier',  
168 CURRENT_TIMESTAMP,  
169 1);  
170 INSERT INTO Alert  
171 (StaffID, Message, CreatedAt, IsRead)  
172 VALUES  
173 (1,  
174 'Flight PK381 delayed due to weather',  
175 CURRENT_TIMESTAMP,  
176 FALSE);  
177 SELECT * FROM Disruption;  
178  
179  
180  
181  
182  
183
```

STDIN
Input for the program (Optional)

Output:

DisruptionID	FlightID	Type	Description
1	1	Pilot Sick	Pilot reported illness before departure